

Search for an Axionlike Dark Matter Candidate Using Nuclear Magnetic Resonance

Abstract

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Axions and axionlike particles (ALPs) are well-motivated dark-matter candidates. Their gradient coupling to nuclear spins produces an oscillating pseudomagnetic field that can be detected using nuclear magnetic resonance (NMR). In this thesis, signal models, frequency-scanning strategies, and data-analysis methods are developed and applied to proof-of-principle searches with the Cosmic Axion Spin Precession Experiment (CASPER).

Two dark-matter scenarios are considered: a virialized Milky Way halo, which produces stochastic signals with finite coherence times, and a halo of ALPs gravitationally bound to Earth, which can produce signals with longer coherence times and a potentially enhanced local density. The frequency-scanning analysis shows that, in the long-measurement limit, coupling sensitivity improves only as $T_{\text{tot}}^{-1/4}$ with total integration time. Longer nuclear-spin coherence times and frequency steps matched to the sensitive bandwidth are therefore central to efficient searches.

The CASPER-Gradient-LowField apparatus combines a thermally polarized liquid-methanol sample with SQUID magnetometry. Searches near 1.35 MHz, corresponding to ALP masses near $5.58 \text{ neV}/c^2$, found no statistically significant candidates. The Milky Way halo search yields a 95 % confidence-level upper limit on the ALP-proton coupling of approximately $|g_{\text{ap}}| \lesssim 1 \times 10^{-2} \text{ GeV}^{-1}$ over a 240 Hz frequency interval. The Earth-bound search yields an approximate 95 % confidence-level limit of $|g_{\text{ap}}| \lesssim 1 \times 10^{-9} \text{ GeV}^{-1}$ over a 48 Hz interval, conditional on the assumed Earth-bound halo density and state population.

These measurements demonstrate liquid-state NMR as a method for searching for ALP-proton gradient coupling under distinct halo assumptions. The very small thermal proton polarization, approximately 1.8×10^{-7} , is the principal limitation on sensitivity. Hyperpolarization offers a route to improvements of several orders of magnitude, complemented by lower detector noise and improved field homogeneity. The signal models, statistical methods, and numerical spin-dynamics simulations developed here provide a basis for future searches with more highly polarized samples.

Supervisor approval

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